

# Evolution: Quick Memory Revision Notes

Key Concepts • Formulas • Evolutionary Chronology • Comparison Tables • Exam Triggers



## 1. ORIGIN OF UNIVERSE & EARTH

- **Big Bang Theory:** Georges Lemaître (1927). Singular massive explosion ~**20 bya**. Universe expanded, temperature dropped; **H** and **He** formed first.
- **Nebular Hypothesis:** Immanuel Kant (1755). Solar system formed ~**4.5–5 bya** from spinning cloud (nebulous).
- **Early Earth:** Formed ~**4.5 bya**. Temperature 4000–8000°C. Density stratification: **Fe, Ni** in Core; **Si, Al** in Crust; **H, C, N, O** in atmosphere.
- **Prebiotic Atmosphere:** Highly **reducing** (no free  $O_2$ ). Composed of  **$CH_4$ ,  $NH_3$ ,  $H_2O$**  vapor, and  **$H_2$** .
- **Origin of Oceans:** UV rays split  $H_2O \rightarrow H_2 + O_2$ ; light  **$H_2$**  escaped. Oxygen oxidized  **$NH_3$  &  $CH_4$** . Rain condensed over millennia to form oceans.
- **First Life:** Appeared ~**4.0 bya** (500 my after Earth's formation).

## 2. THEORIES OF ORIGIN OF LIFE

- **Special Creation:** Father Suarez. World created in 6 natural days. Fixed, immutable species.  
**Scientifically Discarded**
- **Cosmozoic / Panspermia:** Richter (1865). Life arrived as spores (cosmozoa) preserved inside meteorites.  
**Unproven**
- **Spontaneous Generation (Abiogenesis):** Aristotle, Van Helmont. Life arose from non-living matter by "vital force" (e.g., mice from dirty shirt + wheat in 21 days).  
**Disproved**
- **Biogenesis:** Life from pre-existing life (*Omne vivum ex vivo*). Proven by **Francesco Redi** (meat jars), **Spallanzani** (boiled broth), and **Louis Pasteur** (swan-necked flask).

## 3. CHEMICAL EVOLUTION & MILLER'S EXPERIMENT

- **Oparin-Haldane Theory (1924/1929):** Abiogenic origin from inorganic precursors  $\rightarrow$  organic monomers  $\rightarrow$  polymers  $\rightarrow$  protobionts.
- **Miller-Urey Experiment (1953):**
  - **Gases:**  **$CH_4 : NH_3 : H_2 = 2 : 1 : 2$**  plus steam ( **$H_2O$** ).
  - **Condition:** Electric spark at **800°C** for 7 days.
  - **Result:** Synthesized amino acids (**glycine, alanine, aspartic acid**), urea, fatty acids, and purines.
- **Protobionts:**
  - **Coacervates (Oparin, 1938):** Protein-polysaccharide droplets bound by water envelope (no lipid bilayer).
  - **Microspheres (Sydney Fox, 1959):** Heated amino acids polymerize into **proteinoids**; show semi-permeable membrane and budding/fission.

⚡ **Key NEET Trigger:** The shift from reducing to oxidizing atmosphere was driven by oxygenic photoautotrophs (cyanobacteria) releasing  **$O_2$** !

## 4. EVIDENCES FOR EVOLUTION

- **Paleontological:** Fossils in sedimentary rocks. Age determined by radiometric dating ( **$^{14}C$ , K-Ar**).
- **Connecting Links:**
  - **Archaeopteryx:** Reptiles (teeth, claws, long tail)  $\leftrightarrow$  Birds (beak, feathers, wings).
  - **Ichthyostega:** Fishes  $\leftrightarrow$  Amphibians.
  - **Seymouria:** Amphibians  $\leftrightarrow$  Reptiles.
  - **Cynognathus / Therapsids:** Reptiles  $\leftrightarrow$  Mammals.
- **Embryological:** Haeckel's Biogenetic law (*Ontogeny recapitulates phylogeny*) was disproved by **Von Baer** (embryos never pass through adult stages of other animals).
- **Biochemical:** Universal genetic code, conserved ATP, cytochrome c sequences across all taxa.

## 5. HOMOLGY VS. ANALOGY

| Feature    | Homologous Organs                                                                 | Analogous Organs                                                                   |
|------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Origin     | Same basic anatomical structure & origin                                          | Different embryonic origin & design                                                |
| Function   | Adapted for different functions                                                   | Adapted for same/similar functions                                                 |
| Evolution  | Divergent Evolution                                                               | Convergent Evolution                                                               |
| Animal Ex. | Forelimbs of human, cheetah, whale, bat                                           | Wings of butterfly & bird; eyes of octopus & mammal; flippers of penguin & dolphin |
| Plant Ex.  | Thorn of <i>Bougainvillea</i> & tendril of <i>Cucurbita</i> (both modified stems) | Sweet potato (root) & potato (stem)                                                |

## 6. NATURAL SELECTION & SPECIATION

- **Industrial Melanism (*Biston betularia*):**
  - *Pre-industrial (1850s)*: Lichen-covered pale trunks → White-winged moths survived; dark melanic predated.
  - *Post-industrial (1920s)*: Soot-covered dark trunks → Dark moths (*carbonaria*) camouflaged; white predated.
- **Anthropogenic Actions:** Rapid selection of DDT-resistant mosquitoes, antibiotic-resistant bacteria, herbicide-resistant weeds.
- **Types of Natural Selection:**
  - *Stabilizing*: Mean favored, extremes eliminated. Peak narrows (e.g., human birth weight ~3–3.5 kg).
  - *Directional*: One extreme favored. Peak shifts in one direction (e.g., peppered moth, drug resistance).
  - *Disruptive*: Both extremes favored, mean eliminated. **Two peaks formed (bimodal)** (e.g., rabbit fur colour on black/white rocks).

## 7. BIOGEOGRAPHICAL EVIDENCE & ADAPTIVE RADIATION

- **Adaptive Radiation:** Evolutionary divergence of an ancestral form into diverse ecological niches in an isolated area.
  - *Darwin's Finches*: Original mainland seed-eater radiated in Galapagos into cactus-eating, vegetarian bud-eating, and insectivorous woodpecker finches (beak modifications).
  - *Australian Marsupials*: Common marsupial ancestor radiated into diverse forms across the Australian continent.
- **Convergent Evolution (Marsupials vs. Placentals):**

| Placental Mammal | Australian Marsupial Counterpart |
|------------------|----------------------------------|
| Mole             | Marsupial mole                   |
| Anteater         | Numbat (banded anteater)         |
| Mouse            | Marsupial mouse                  |
| Lemur            | Spotted cuscus                   |
| Flying squirrel  | Flying phalanger (sugar glider)  |
| Bobcat           | Tasmanian tiger cat              |
| Wolf             | Tasmanian wolf (Thylacine)       |

## 8. EVOLUTIONARY THEORIES: CORE COMPARISON

| Parameter                | Lamarckism                        | Darwinism                                  | Mutation Theory                                 |
|--------------------------|-----------------------------------|--------------------------------------------|-------------------------------------------------|
| <b>Proponent</b>         | J.B. Lamarck (1809)               | Charles Darwin (1859)                      | Hugo de Vries (1901)                            |
| <b>Mechanism</b>         | Use and disuse; acquired traits   | Natural selection & branching descent      | Single-step large mutation ( <b>Saltation</b> ) |
| <b>Variations</b>        | Somatic & directional             | Continuous, small, directional             | Discontinuous, sudden, random                   |
| <b>Disproof / Status</b> | Disproved by Weismann (Germplasm) | Accepted basis; lacks origin of variations | Integrated into Modern Synthetic Theory         |

## 9. HARDY-WEINBERG PRINCIPLE

Allele frequencies remain constant in a large, randomly mating population over generations without evolutionary disturbances.

**Allele Frequencies:**  $p + q = 1$  | **Genotypic:**  $p^2 + 2pq + q^2 = 1$   
 ( $p^2 = AA$ ,  $2pq = Aa$ ,  $q^2 = aa$ )

- **5 Disruptive Factors:** (1) Gene migration / flow, (2) Genetic drift (Bottle-neck & Founder effect), (3) Mutation, (4) Genetic recombination, (5) Natural selection.

## 10. GEOLOGICAL TIME SCALE SNAPSHOT

- **Eras (Oldest to Recent):** Archaeozoic → Proterozoic → Paleozoic → Mesozoic → Cenozoic.
- **Key Milestones:**
  - *Paleozoic Era:* Age of Invertebrates (Cambrian/ Ordovician), Age of Fishes (Devonian), Age of Amphibians (Carboniferous - coal forests).
  - *Mesozoic Era:* **Golden Age of Reptiles & Gymnosperms.** Triassic (first dinosaurs) → Jurassic (dinosaurs rule, Cycads dominant, toothed birds appear) → Cretaceous (extinction of dinosaurs).
  - *Cenozoic Era:* **Age of Mammals & Angiosperms.** Tertiary + Quaternary (Holocene: recent, human civilization).
  - *Plant Ancestry:* *Psilophyton* is common ancestor of horsetails, ferns, and gymnosperms.

## 11. HOMINID EVOLUTION: CHRONOLOGY & CRANIAL CAPACITY

| Ancestor                    | Time / Epoch        | Cranial Capacity  | Crucial Hallmarks                                                           |
|-----------------------------|---------------------|-------------------|-----------------------------------------------------------------------------|
| <b>Dryopithecus</b>         | 15–20 mya (Miocene) | ~350–400 cc       | Arboreal, knuckle-walker, arms=legs, ape-like ancestor.                     |
| <b>Ramapithecus</b>         | 14–15 mya (Miocene) | ~400–450 cc       | Siwalik Hills (India), parabolic jaw, upright walking, more man-like.       |
| <b>Australopithecus</b>     | 2–4 mya (Pliocene)  | 450–600 cc        | First bipedal hominid ("Lucy", "Taung Baby"), stone weapons, fruit eater.   |
| <b>Homo habilis</b>         | 1.6–2.0 mya         | <b>650–800 cc</b> | "Handy man", first tool maker (Oldowan tools), did not eat meat.            |
| <b>Homo erectus</b>         | ~1.5 mya            | <b>~900 cc</b>    | Java/Peking man, fully erect posture, used fire & Acheulean axes, ate meat. |
| <b>Neanderthal man</b>      | 100,000–40,000 YA   | <b>~1400 cc</b>   | Cave dwellers, heavy brow ridge, used hides, buried dead with flowers.      |
| <b>Cro-Magnon</b>           | 34,000–10,000 YA    | <b>~1650 cc</b>   | Direct modern ancestor, tall, prominent chin, cave paintings (Lascaux).     |
| <b>Homo sapiens sapiens</b> | 75,000 YA–present   | 1350–1450 cc      | Arose in Africa during ice age; agriculture & settlements ~10,000 YA.       |

 **Skull Allometry Trigger:** The skull of a baby chimpanzee resembles an adult human skull more closely than an adult chimpanzee skull!